

Helium Ions versus Iodine 125 Brachytherapy in the Management of Uveal Melanoma

A Prospective, Randomized, Dynamically Balanced Trial

Devron H. Char, MD,¹ Jeanne M. Quivey, MD,² Joseph R. Castro, MD,³ Stewart Kroll, MS,¹ Theodore Phillips, MD²

Purpose: Optimal radiation therapy for uveal melanoma is uncertain, and the relative efficacies of radioactive plaques and charged particles are unclear.

Methods: The authors prospectively studied helium-ion irradiation and iodine 125 (¹²⁵I) brachytherapy in a randomized, dynamically balanced trial. Of the 184 patients who met the eligibility criteria, 86 were treated with helium ions and 98 with ¹²⁵I brachytherapy.

Results: No patients with uveal melanoma had a history of systemic malignancy. Tumors were less than 15 mm in maximum diameter and less than 10 mm in thickness. A minimum tumor dose of 70 GyE was delivered to the tumor apex. There was a significantly higher local recurrence rate after ¹²⁵I brachytherapy than after helium-ion irradiation. Enucleations occurred more frequently after brachytherapy (relative risk = 1.99; 95% confidence interval, 0.78–5.78). More anterior segment complications occurred after helium-ion irradiation. To date, there has been no measurable impact on survival.

Conclusions: Most uveal melanomas can be managed with radiation with retention of the eye. There was better tumor control with helium-ion irradiation; however, there were more anterior segment complications. *Ophthalmology* 1993;100:1547–1554

Radiation therapy has been used in uveal melanoma management since 1929; however, a number of issues

regarding its use remain unresolved.¹ The relative advantages and disadvantages of charged particles versus brachytherapy are uncertain.²

Ocular brachytherapy with radon seeds was first reported in 1930; radioactive plaques have been made with various isotopes initially to treat mainly small or medium tumors or melanomas in one-eyed patients.^{3,4} Most early radioactive plaques were constructed with cobalt 60 (⁶⁰Co); however, significant ocular morbidity was noted with this high-energy gamma emitter.^{5,6} When isodose curves were plotted, to treat a medium or large posterior choroidal melanoma, visually vital

Originally received: November 4, 1992.

Revision accepted: March 26, 1993.

¹ Department of Ophthalmology, University of California, San Francisco.

² Department of Radiation Oncology, University of California, San Francisco.

³ Lawrence Berkeley Laboratory, University of California, Berkeley.

Presented at the American Academy of Ophthalmology Annual Meeting, Dallas, November 1992.

Supported in part by NIH grant EYO 7504, NCI CA 19138 DOE-DE-ACO3-765F00098, Bethesda, Maryland, and by unrestricted grants from That Man May See, San Francisco, California, and Research to Prevent Blindness Inc, New York, New York.

Reprint requests to Devron H. Char, MD, Ocular Oncology Unit, 10 Kirkham, Box 0730, University of California, San Francisco, CA 94143.